

Coding & Solution

Date	29 June 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Coding & Solution

This section presents the implementation of the Credit Card Approval Prediction System. The solution uses Machine Learning to predict whether a credit card application is likely to be approved or rejected based on applicant information. The application is developed using Python, Flask, HTML, and CSS, following clean coding practices, modular design, and project requirements to ensure reliability, maintainability, and ease of future enhancements.

Instructions:

1. Ensure the code is well-structured, commented, and follows best coding practices.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, trained model files, and setup instructions required to run the application.
4. Attach or link to the GitHub repository in the table below.

Solution Summary

Field	Details
Repository Link / URL	https://github.com/Keerthireddy0103/CreditCardApprovalPrediction
Programming Language(s)	Python, HTML, CSS
Framework(s) Used	Flask, Scikit-learn
Key Features Implemented	Credit card approval prediction, data analysis, data preprocessing, feature encoding, Machine Learning model training, Random Forest model integration, responsive web interface.
Pending / Incomplete Features	User authentication, database integration, prediction history, email notifications.
Setup / Run Instructions	Install the required dependencies using <code>pip install -r requirements.txt</code> , then run <code>python app.py</code> and open the application in a web browser.

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into separate modules/files	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments/documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a GitHub repository	Yes

Additional Notes / Comments:

The Credit Card Approval Prediction System successfully integrates a Random Forest Machine Learning model with a Flask web application to provide accurate credit card approval predictions. The project follows a modular architecture with separate folders for source code, templates, static files, datasets, and trained models, making it easy to maintain and extend. The application has been successfully deployed on Render, and the complete source code is available on GitHub for version control and future development.